Sam Rosen

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Education

Duke University **May 2026**
Ph.D., Statistical Science; **Advisor:** Jason Xu Durham, NC
Dissertation Title: "Convex Optimization Methods for Structured Statistical Problems"

- Summer Research Fellowship (2025)
- Honorable Mention for Teaching Assistant of the Year (2023)
- Bass Connections Fellowship (2022)

University of Massachusetts: Amherst **May 2021**
B.S., Computer Science; **B.S.**, Mathematics *(3.9/4.0 GPA)* Amherst, MA

- Recipient of UMass Chancellor's Award – Four-year academic scholarship

Professional Experience

Pinterest **Summer 2024**
Machine Learning Engineering Intern San Francisco, CA

- Trained and tuned deep learning models for content-to-content recommendation systems.
- Conducted successful A/B testing to show efficacy of changes on production systems.

Cruise LLC (General Motors) **Summer 2022**
Machine Learning Acceleration Intern San Francisco, CA

- Implemented robust and nonparametric standard error calculations to internal simulation metrics.
- Added Kernel Density Estimation/Bandwidth Selection to automatically analyze performance degradation.

Genentech **Summer 2021**
Research and Development Intern San Francisco, CA

- Lead development of *epiviz.gl*, a JS framework for visualizing genomic data with WebWorkers and WebGL, capable of rendering millions of data points and entire chromosomes at 60 FPS with high precision.
- Developed data selection, rendering, navigation with a pseudo grammar-of-graphics implementation.

Datadog **Sept. 2020 – Dec. 2020**
Software Engineering Co-op (Cloud Integrations Team) New York, NY

- Optimized and bolstered Azure crawlers responsible for crawling millions of data points an hour.
- Debugged and implemented fixes for issues found by customers in production for crawled metrics.

DraftKings **Summer 2019**
Software Engineering Intern (DevOps Team) Boston, MA

- Created a scalable application for live tracking of release branches to production using AWS Lambda.
- Designed serverless architecture scalable to arbitrary codebase size with complete up-to-date release data.
- Designed DynamoDB schema and frontend with React for a responsive, efficient API and user interface.

Johns Hopkins University: Applied Physics Lab **Summer 2017, 2018**
Software Engineering Intern (Large-Scale Analytics Group) Laurel, MD

- Programmed low-memory implementations of machine learning algorithms for training on arbitrarily large data.
- Created analytics for graph multi-edge merging, time series, and data fusion using Java and MapReduce.
- Developed a random forest algorithm on a distributed data system for classifying attributes on graph vertices.

Publications

Sam Rosen and Jason Xu (Aug. 2026). "Constrained Weighted Bayesian Bootstrap". In: *Conference on Uncertainty in Artificial Intelligence* 42, Accepted and to appear. arXiv: 2606.04237 [stat.ME]. PREPRINT: https://arxiv.org/abs/2606.04237.

Sam Rosen and Jason Xu (May 2026). "Affinity Graph Connectivity in Convex Clustering". Submitted to *SIAM Journal on Mathematics of Data Science*. arXiv: 2605.24673 [stat.ML]. PREPRINT: https://arxiv.org/abs/2605.24673.

Sam Rosen, Eric C. Chi, and Jason Xu (Apr. 2026). "Biconvex Biclustering". Submitted to *Journal of the American Statistical Association*. arXiv: 2604.03936 [stat.ML]. PREPRINT: https://arxiv.org/abs/2604.03936.

Jiachang Liu, **Sam Rosen**, Chudi Zhong, and Cynthia Rudin (Dec. 2023). "OKRidge: Scalable Optimal k-Sparse Ridge Regression". In: *Advances in neural information processing systems* 36. NeurIPS 2023 Spotlight Paper, pp. 41076–41258.

Skills, Interests and Service

Programming Languages: Python***, R***, Java***, JavaScript***, C/C++**, Julia**, Scala*, Matlab*
Research Interests: Statistical Computing, Convex Optimization, Clustering, High-performance Computing
Reviewer: AAAI 2026 "Model Uncertainty and the Rashomon Effect" Workshop
Teaching: TA for Graduate Statistical Computing (R and Python), Regression Analysis, Data Visualization; Instructor for Statistical Science Master's Python Bootcamp (2023, 2024, 2025)